

```
import pandas as pd

file_path = '/content/dataset_youtube-scraper_2026-02-09_08-50-53-934.xlsx'
df = pd.read_excel(file_path)

print("Excel file loaded successfully. Displaying the first 5 rows:")
df.head()
```

Excel file loaded successfully. Displaying the first 5 rows:

	channelId	channelName	channelUrl	channelUsername	collaborators	commentsCount
0	UCMSI1Ck1mJOaxwJ0bzyhQ	The Urban Fight	https://www.youtube.com/@TheUrbanFight	TheUrbanFight	NaN	10596

1 rows × 55 columns
Warning: Total number of columns (55) exceeds max_columns (20) limiting to first (20) columns.

```
print("First 5 rows of the DataFrame:")
df.head()

print("\nDescriptive statistics of the DataFrame:")
df.describe()
```

First 5 rows of the DataFrame:

Descriptive statistics of the DataFrame:

	commentsCount	duration	likes	numberOfSubscribers	viewCount
count	1.0	1.0	1.0	1.0	1.0
mean	10596.0	468.0	1200000.0	3390000.0	23721082.0
std	NaN	NaN	NaN	NaN	NaN
min	10596.0	468.0	1200000.0	3390000.0	23721082.0
25%	10596.0	468.0	1200000.0	3390000.0	23721082.0
50%	10596.0	468.0	1200000.0	3390000.0	23721082.0
75%	10596.0	468.0	1200000.0	3390000.0	23721082.0
max	10596.0	468.0	1200000.0	3390000.0	23721082.0

```
print("Identifying columns with all missing values...")

# Identify columns where all values are missing
missing_cols = df.columns[df.isnull().all()].tolist()

if missing_cols:
    print(f"Columns with all missing values to be dropped: {missing_cols}")
    # Drop these columns from the DataFrame
    df = df.drop(columns=missing_cols)
    print("Columns dropped successfully.")
else:
    print("No columns found with all missing values.")
```

Identifying columns with all missing values...
No columns found with all missing values.

```
print("Checking for duplicate rows...")

duplicate_rows_count = df.duplicated().sum()

if duplicate_rows_count > 0:
    print(f"Found {duplicate_rows_count} duplicate rows. Removing them...")
    df.drop_duplicates(inplace=True)
    print("Duplicate rows removed successfully.")
else:
    print("No duplicate rows found.")
```

```
print(f"\nDataFrame shape after handling duplicates: {df.shape}")
```

```
Checking for duplicate rows...
No duplicate rows found.
```

```
DataFrame shape after handling duplicates: (1, 49)
```

```
print("Converting 'date' column to datetime...")
df['date'] = pd.to_datetime(df['date'])
print("'date' column converted successfully.")
```

```
print("\nInspecting 'duration' column for potential conversion:")
print(df['duration'].head())
```

```
Converting 'date' column to datetime...
'date' column converted successfully.
```

```
Inspecting 'duration' column for potential conversion:
0      468
Name: duration, dtype: int64
```

```
print("Displaying head of the cleaned DataFrame:")
df.head()
```

```
print("\nDisplaying descriptive statistics of the cleaned DataFrame:")
df.describe()
```

Displaying head of the cleaned DataFrame:

Displaying descriptive statistics of the cleaned DataFrame:

	commentsCount	duration	likes	numberOfSubscribers	viewCount	
count	1.0	1.0	1.0	1.0	1.0	
mean	10596.0	468.0	1200000.0	3390000.0	23721082.0	
std	NaN	NaN	NaN	NaN	NaN	
min	10596.0	468.0	1200000.0	3390000.0	23721082.0	
25%	10596.0	468.0	1200000.0	3390000.0	23721082.0	
50%	10596.0	468.0	1200000.0	3390000.0	23721082.0	
75%	10596.0	468.0	1200000.0	3390000.0	23721082.0	
max	10596.0	468.0	1200000.0	3390000.0	23721082.0	

```
print("Converting 'duration' column to total seconds...")
```

```
def duration_to_seconds(duration_str):
    if pd.isna(duration_str):
        return None
    # Assuming format 'HH:MM:SS'
    parts = str(duration_str).split(':')
    if len(parts) == 3:
        h, m, s = map(int, parts)
        return h * 3600 + m * 60 + s
    elif len(parts) == 2: # MM:SS format might occur
        m, s = map(int, parts)
        return m * 60 + s
    return None # Handle other unexpected formats if any

df['duration'] = df['duration'].apply(duration_to_seconds)

print("'duration' column converted successfully. Displaying first 5 rows of the converted column:")
print(df['duration'].head())
```

```
Converting 'duration' column to total seconds...
'duration' column converted successfully. Displaying first 5 rows of the converted column:
0      None
Name: duration, dtype: object
```

